Antoine KRAUS

AI Engineer
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Summary

Vice-President and co-founder of the Computer Science Research Association.

Passionate Junior AI Engineer (<3 years). My experience with **NixOS** has instilled a rigorous discipline for reproducibility, precision, and deep debugging across multiple languages.

Education

School Georges Cabanis Brive-la-Gaillarde • 2020 – 2022
Intensive STEM Program, France
Specialization: Intensive two-year undergraduate program preparing for competitive entrance examinations to top-tier French engineering schools, with a heavy focus on Advanced Mathematics and Physics (Maths Sup/Spé).

Experience

Summer jobs France
Material Handling [2019] | Temp Work [2021 - 2022 - 2023] | Teillet Farm [2024 - 2025]

Maison de l'Intelligence Artificielle Morocco - Oujda
04/2025 – 08/2025

Computer Vision Model Training: Trained and fine-tuned computer vision models (YOLOv8, YOLOv11, RT-DETR, D-Fine) on multiple custom datasets based on specific use cases.

Embedded System Development: Developed an embedded system utilizing OpenCV and OpenVINO for deployment optimization.

CAD Design & 3D Printing: Designed and 3D-printed a tricopter drone with a 1.5-meter wingspan.

SPIE Turbomachinery Pau, France
Project Manager & Software Engineer 09/2025 – 09/2026
Requirements Engineering & R&D: Reviewed technical specifications and benchmarked open-source ERP solutions to identify optimal foundational architectures.

Full-Stack Development: Designed and engineered a custom Computer-Aided Logistics Management system from scratch using **Python** and **JavaScript**, backed by **MariaDB**.

Reproducible Infrastructure: Leveraged **NixOS** and **WSL** to establish a highly reliable, deterministic development environment, ensuring absolute consistency from local machines to production.

Containerization & Cloud Deployment: Containerized the application ecosystem with **Docker** and successfully deployed and orchestrated the infrastructure on **Azure**.

Project Ownership: Managed the full software development lifecycle (SDLC), bridging the gap between operational requirements and robust, scalable deployment.

Projects

Data Parser & Predictive AI Agent 04/2025
Data Parsing & Telemetry: Developed a high-performance data parser in **C** to process raw game logs and extract real-time state features.

Predictive Modeling: Engineered and trained an AI agent in **Python** designed to predict match outcomes based on live in-game telemetry.

Next-Gen Architecture: Designing a predictive problem-solving engine aimed at forecasting a continuous sequence of future events over a given time horizon, starting from any arbitrary game state.

ML-Driven Vulnerability Detection Engine 03-2026 05-2026
Vulnerability Assessment: Engineered a machine learning system to detect security vulnerabilities within the LLVM Intermediate Representation (IR), covering source code, build systems, and the software supply chain.

High-Performance Extraction: Developed a custom, high-speed LLVM IR feature extractor written in **Rust** to ensure memory safety and fast static analysis processing.

Deep Learning Pipeline: Built and trained the predictive vulnerability model in **Python** using extracted IR representations.

Languages

French
Native

English
Fluent

Interests

Climbing **Hiking** **Cooking**